

In vanilla backpropagation,

$$\min_{\Theta} \sum_{t=1}^T \sum_{i=1}^n \text{loss}(f_{\Theta}(v_{ti}), y_{ti})$$

where, t is task id, i is the datapoint in the task, Θ is the neural net parameter
 v_{ti} is the input to the model, y_{ti} is the true label

In Query based and Attentive task,

$$\min_{\Theta} \sum_{t=1}^T \sum_{i=1}^n \text{loss}(f_{\Theta, B_t \sim \phi_t}(v_{ti}), y_{ti})$$

where, B_t is the buffer filled with sampled elements from task t .

ϕ_t is the fixed distⁿ of the data at task t

For simplicity, let's assume all the elements from the buffer is being used to train the model.

We can then write this as,

$$= \min_{\Theta} \sum_{t=1}^T \sum_{i=1}^n \text{loss} \left(f_{\Theta} (q_{ti}, s_{t1}, s_{t2}, \dots, s_{t|B|}), y_{ti} \right)$$

$|B| \Rightarrow \text{size of buffer}$

$$= \min_{\Theta} \sum_{t=1}^T \sum_{i=1}^n \text{loss} \left(f_{\Theta} (q_{ti}, s_{t1}) \odot \hat{V}_1 \right. \\ \left. + f_{\Theta} (q_{ti}, s_{t2}) \cdot \hat{V}_2 \dots \right), y_{ti} \Big)$$

$$= \min_{\Theta} \sum_{t=1}^T \sum_{i=1}^n \text{loss} \left(V \cdot f_{\Theta} (q_t, s_t) \right), y_{ti} \Big)$$

where, V is a matrix whose value is based on either attention network or query based

$$\Theta'_t = f_{\Theta} (q_t, s_t) = [\Theta'_1, \Theta'_2, \dots, \Theta'_B]_t$$

Now, Θ_t captures the property of a task in terms of relationship between query and supports in that task. Thus, Θ_t represents the approximate information of a task t .

If Θ'_t is approximately constant this means a specific pattern exists across tasks which can be learnt under certain ~~and~~ assumptions.

→ The rationale is that in the usual learning of stationary dist'n, the assumpt'n is datapoints in the dist'n have a specific underlying pattern.

→ In above scenario, each datapoint is replaced by complete tasks and a common pattern across tasks is captured in Θ'_t

→ The common pattern is defined by the metric learning constraint,

$$|\Theta'_{\text{query}} - \Theta'_{\text{matching support}}|_t > |\Theta'_{\text{query}} - \Theta'_{\text{non-matching support}}|_t$$

for $\forall t$

The above constraint is same for all the tasks i.e. for all tasks we expect same behaviour captured in Θ'_t

To provide further evidence, we try in parallel with another algorithm that generates θ'_t for tasks but need not necessarily have θ'_t be a constant across all t . In model agnostic meta learning we have,

$$\theta'_t = \theta - \frac{\partial l(\theta(x_t), y_t)}{\partial \theta}$$

$$\theta = \theta - \sum_{t=1}^T \frac{\partial l(\theta(x_t), y_t)}{\partial \theta'} \cdot \frac{\partial \theta'_t}{\partial \theta}$$

Here, $\therefore \theta'_t$ need not be constant across tasks thus producing suboptimal solution in continual learning setting.

Hence, in MAML we expect to learn θ from just training across tasks and not from a stream of tasks as in continual learning.

In our continual learning scenario we expect infinite stream of data all following a common pattern captured in θ'_t . Thus, we can say that learning the task is similar to learning individual datapoints sampled randomly.